

1. Calculate

(a) $\int_0^\pi \sqrt{1 + \cos x} dx$. Hint: $\cos x = 2 \cos^2 \frac{1}{2}x - 1$.

(b) $\int_1^2 \frac{e^{1/x}}{x^2} dx$.

(c) $\int_0^{\pi/4} \frac{\arctan x}{1+x^2} dx$.

(d) $\int \frac{dx}{x^2 - 4x + 13}$.

(e) $\int \frac{1}{x \ln x} dx$.

(f) $\int_{-1}^1 \frac{x^2}{1+x^2} dx$.

(g) $\int_0^{1/2} \frac{1+x}{\sqrt{1-x^2}} dx$.

(h) $\int \frac{x}{1-x^4} dx$.

(i) $\int_1^c \frac{\ln x^3}{x} dx$.

(j) $\int a^x e^x dx$.

(k) $\int_0^2 x 2^x dx$.

(l) $\int x^2 (2x - 1)^{-7} dx$.

(m) $\int_0^1 \ln(1 + x^2) dx$.

(n) $\int x^n \ln x dx$, $n \neq -1$.

(o) $\int x^3 \sin x dx$.

(p) $\int \cos \sqrt{x} dx$.

(q) $\int \frac{1}{x} \arcsin(\ln x) dx$.

(r) $\int \cos(\ln x) dx$.